CareerOS

AI-Powered Career Operating System

Connecting Students · Universities · Employers

PRODUCT DOCUMENTATION

Team GongBaiWan · Phase 2 Prototype · July 2026

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1. Executive Summary

CareerOS is an AI-first career operating system that connects student evidence, university programme intelligence, and employer talent workflows through a shared intelligence layer called CareerGraph. Built for the higher education and early talent ecosystem, CareerOS addresses a fundamental breakdown in how students prove readiness, how universities understand market alignment, and how employers evaluate early-career talent.

Phase 2 evolves CareerOS from a dashboard-first prototype into an AI-first Operating System. The core principle: **"Most career platforms organise information. CareerOS organises action."** Every workspace is now led by an AI companion that surfaces what matters, proposes the next step, and routes the user directly to the right action — removing the need to navigate menus or interpret dashboards.

The platform delivers three purpose-built workspaces — Student, University, and Employer — each powered by AI assistants and unified by a common evidence and intelligence layer. Rather than operating as isolated tools, these workspaces reinforce one another: student evidence improves university intelligence, university interventions raise graduate readiness, and employer engagement feeds outcome data back into the system.

2. Problem Statement

The career readiness ecosystem is broken across three interconnected stakeholders. The scale of the problem is significant:

85%	$2.4T	72%
Graduates underqualified at point of hiring	Global skills mismatch cost annual productivity loss	Employers struggle to evaluate early talent from resumes
TalentBank Malaysia 2024	WEF Future of Jobs 2025	LinkedIn Talent Trends 2025

Key market statistics driving the CareerOS problem statement

2.1 Students / Candidates

- Cannot prove readiness beyond a resume — skills are self-reported, unverified, and disconnected from real evidence.
- Face too many generic opportunities with no intelligent filtering based on their actual profile.
- Lack clear, personalised guidance on which skills to build and which career paths match their evidence.
- Track applications across scattered tools with no unified pipeline.
- Discover skill gaps only at the point of rejection — too late to act on them.

2.2 Universities

- Curriculum cycles are slow and disconnected from real-time employer skill demand.
- Cannot see at-risk cohorts or skill gaps across programmes until outcomes are already poor.
- Event impact — from hackathons to career fairs — is rarely measured or used in decision-making.
- Market data and student evidence are fragmented across systems with no shared intelligence layer.
- Accreditation preparation (MQA, SETARA, QS) is episodic, manual, and evidence is repeatedly recollected.

2.3 Employers

- Resume screening is noisy — early talent is hard to evaluate on keyword-matched documents alone.
- Campus engagement is fragmented with no structured way to create, run, and measure student programmes.
- Applicant pipelines do not connect to verified skill evidence, creating poor hiring signal.
- Strong candidates encountered through events disappear into spreadsheets and are forgotten.

These three broken relationships compound each other. Students graduate underprepared because universities lack market signal. Employers hire inefficiently because candidates cannot prove readiness. Universities cannot improve without outcome data. CareerOS closes this loop.

3. Target Users

CareerOS serves three stakeholder groups, each with a dedicated workspace and representative demo persona.

3.1 Students / Candidates

Demo persona: Chris Lee — Y3 Data Science student, Taylor's University

Who they are: Students preparing for internships, graduate roles, and early career decisions — particularly Malaysian Gen Y and Gen Z graduates entering competitive knowledge economy roles.

Primary goals:

- Understand their career readiness and identify skill gaps before graduation.
- Build a Career Memory — an evidence-backed profile from real experiences, projects, and activities.
- Find relevant events, internships, and jobs matched to their actual skills.
- Track job applications through a structured pipeline.
- Practice interviews and improve professional communication in a low-risk environment.
- Connect with communities and mentors for guidance and peer support.

3.2 Employers / Industry Partners

Demo persona: Edwin Khoo — HR Manager, Google Malaysia

Who they are: Recruiters, HR managers, university relations teams, and industry partners at companies actively hiring early talent from Malaysian universities.

Primary goals:

- Discover evidence-backed early talent without relying on keyword-filtered resumes.
- Understand candidate fit quickly through AI-generated match explanations.
- Create structured student engagement programmes to build talent pipelines.
- Manage job postings and move applicants through a structured hiring pipeline.
- Reactivate strong candidates from past events and challenges before searching cold.

3.3 Universities

Demo persona: Dr. Evelyn Chen — Dean of Computing & AI, Heriot-Watt University Malaysia

Who they are: Deans, programme directors, career teams, curriculum committees, and university leaders responsible for graduate outcomes and programme quality.

Primary goals:

- Understand programme-market alignment and identify curriculum gaps.
- Detect student readiness risks by cohort before graduation.
- Measure the impact of events, collaborations, and interventions.
- Use alumni outcome data to inform curriculum and collaboration decisions.
- Prepare for accreditation (MQA, SETARA, QS) continuously — not in deadline panic.
- Coordinate institutional decisions through AI-assisted multi-function briefings.

4. Product Positioning

CareerOS is a career intelligence platform for higher education and early talent ecosystems. It is not a resume builder, a generic job board, a LinkedIn clone, a generic LMS, or a university admin dashboard. It is an AI-first operating system that turns fragmented career activity into connected, actionable intelligence.

4.1 Positioning Statement

CareerOS is the only platform that turns fragmented career activity into connected intelligence — giving students verified career evidence, universities programme-market insight, and employers transparent talent discovery — all within a single shared ecosystem. In Phase 2, the AI Companion becomes the primary interface: users start with intent, not menus.

4.2 How CareerOS Differs from Existing Solutions

Competitor	Their Focus	CareerOS Difference
LinkedIn	Public professional networking	Verified evidence and readiness, not public profiles
Handshake	Campus job listings and recruiting	Adds student memory, programme intelligence, post-event impact
Indeed	High-volume job search aggregation	Career preparation and evidence building, not listing volume
Glassdoor	Company reviews and salary transparency	Forward-looking readiness intelligence, not backward-looking reviews
JobStreet	SEA job listings	Connects preparation, evidence, matching, and outcomes
Hiredly	Malaysian Gen Y/Z job board with culture fit	Evidence over personality matching; university as stakeholder
LMS Platforms	Course content delivery for institutions	Employability outcomes, market alignment, employer connection

4.3 Core Differentiators

- Evidence-first profiles — skills backed by verified experiences, not self-reported keywords.
- CareerGraph intelligence layer — shared across students, universities, and employers.
- Three-sided ecosystem — all three stakeholders in one connected system.
- Programme-market alignment — showing universities exactly where curriculum lags demand.
- Post-event impact measurement — turning university collaboration activity into strategic intelligence.
- Match explainability — employers see why a candidate fits, not just a percentage score.
- AI-first navigation — natural language input routes users to the right action, eliminating menu friction.

- University AI Office — specialised AI roles and a human-governed Decision Room for institutional leadership.

5. Solution Overview

CareerOS connects student evidence, university intelligence, and employer talent workflows through CareerGraph — a shared intelligence layer that links skills, experiences, opportunities, employers, university outcomes, and AI recommendations across the platform.

5.1 CareerGraph

CareerGraph is the product's core intelligence layer. It powers skill extraction from student experiences, career path recommendations, programme-market alignment analysis, employer candidate matching, and cross-side outcome feedback. In Phase 2, CareerGraph is extended to support NLP-driven navigation — users describe their intent in natural language and CareerGraph routes them to the correct workspace, module, and pre-filled action.

5.2 Three-Sided Ecosystem

- **Student:** builds evidence, understands readiness, finds opportunities, tracks applications, practises interviews.
- **University:** monitors readiness, aligns programmes to market demand, measures event and collaboration impact, manages accreditation evidence, coordinates institutional decisions via AI Office.
- **Employer:** discovers verified talent, creates student engagement programmes, manages applicant pipelines, reactivates warm candidates.

5.3 Cross-Side Flywheel

Student evidence improves university intelligence. University interventions raise graduate readiness. Employer engagement and event outcomes feed back into student guidance and university decisions. This flywheel makes CareerOS a self-improving system — not a static directory or job board.

Student Builds evidence Finds opportunities Tracks applications Practises interviews	**CareerGraph** Intelligence Layer Evidence · Skills Paths · Matching Insights · Actions	**Employer** Discovers talent Creates programmes Hires verified Reactivates warm talent
	University Monitors readiness Aligns curriculum Measures impact AI Office decisions	

The CareerOS three-sided flywheel — value grows as all stakeholders participate

6. Product Walkthrough

CareerOS exposes three purpose-built workspaces. Each has its own sidebar navigation, AI companion, and role-specific data. Phase 2 introduces NLP navigation, an AI-first companion interface, and a significantly enhanced University workspace with specialised AI agents and human-governed decision infrastructure.

6.1 Student Workspace

Persona: Chris Lee — Y3 Data Science student, Taylor's University. Accessible at /student.

Overview Dashboard

AI-first command centre. The page opens with a robot AI companion delivering a proactive briefing: what changed, what matters, and what to do next. Three signal cards surface inside a conversation bubble (e.g., a deadline alert, a readiness gap, a strong match). A Footprint section shows unfinished actions from previous sessions. Quick reply chips below the robot let users respond with one tap rather than typing. The Career Readiness Score gauge, Monthly Career Actions checklist, and curated recommendations are available on demand without cluttering the primary view.

AI Companion (NLP Navigation)

The primary interface in Phase 2. The user types or speaks their intent in natural language (e.g., "I want to practice for my Google interview" or "show me GenAI jobs"). The companion sends the intent to the Groq LLM (Llama 3), which classifies it and returns a structured JSON response: { intent, destination, action, prefill }. The frontend reads this and navigates directly to the correct page, pre-filling forms or filters where appropriate. This eliminates the need for users to know the menu structure.

Career Memory

The student's living career record — the source of truth for skill evidence. Phase 2 replaces the static form with a conversational add-experience flow: (1) the user selects an experience type from four choice cards (internship, project, leadership, activity); (2) the AI robot asks targeted questions and the user responds naturally; (3) the Groq LLM extracts skills from the user's text (real inference, not hardcoded); (4) a confirmation card shows detected skills with evidence source. The Experience Timeline, extracted Top Skills with frequency counts, Career Momentum trend signals, Career Themes, and AI-powered Resume Panel with PDF download are all available in this module.

Career Intelligence

AI-powered analytical insights about the student's skills and career position. Tabbed across Skills, Career Path, and Market Insights. Shows Skill Summary Cards, Skill Clusters by category (Technical, Business, Leadership, Communication), Improvement Opportunities ranked by demand, and live labour market data per role and skill. The Career Graph displays related roles with suitability, demand, and salary range — reducing degree lock-in by surfacing adjacent career paths.

Opportunities

Event and job discovery hub. Event Hub tab shows match-scored event cards filterable by category (Hackathons, Case Competitions, Workshops, Talks, Webinars, Networking) with a Deadline Alert Banner for urgent opportunities. Job Market and Saved tabs complete the hub. All opportunities are ranked by AI match percentage against the student's current profile.

Applications

Full application pipeline tracker. KPI tiles show Total Applications, Interview Rate, Response Rate, and Average Match Score. A Kanban board organises applications across Applied, Under Review, Interview, Assessment, and Offer stages.

Interview Practice

Repeatable mock interview sessions linked to active applications and the student's evidence. The AI companion provides role-specific questions, voice practice support, and response feedback — helping students improve articulation in a low-risk environment before the real opportunity.

Communities & Mentorships

Path-recommended communities, interview stories, portfolio/resume feedback, career-switch discussions, teammate finding, and mentorship discovery. Communities are the platform's retention mechanism after rejection — giving students a constructive reason to return even when applications are not progressing.

6.2 University Workspace

Persona: Dr. Evelyn Chen — Dean of Computing & AI, Heriot-Watt University Malaysia. Accessible at /university.

Overview — Career Intelligence Hub

Central command for programme performance. KPI tiles show Market Alignment Score, Graduate Readiness Score, Students at Risk count, and Top Emerging Gap. Includes an AI University Advisor insight block and one-click export. All KPIs update cross-page as actions are taken (e.g., completing interventions decrements the at-risk count via shared Zustand state).

University AI Office

The flagship Phase 2 feature. A hub of five specialised AI agents — each with a distinct institutional role — that monitor their domain continuously and surface briefings, alerts, and recommended decisions to the Dean. The five agents are:

- **Dr. Atlas** (Chief Academic Intelligence) — curriculum gaps, market alignment, accreditation priorities.
- **Prof. Nova** (Student Success Officer) — cohort readiness risks, intervention queue, welfare signals.
- **Dean Echo** (Strategic Partnerships) — collaboration health, employer engagement, new partner recommendations.
- **Dr. Iris** (Graduate Outcomes & Alumni) — salary trends, role pathways, alumni feedback routing.
- **Aria** (Compliance & Accreditation) — MQA/SETARA/QS requirement status, evidence ownership, deadline tracking.

Each agent card displays department-specific update bullets with urgency indicators. Clicking "Enter Decision Room" opens a modal where the agent delivers a prepared briefing, presents a recommended action with trade-offs, and asks for human approval. All five agents attend every meeting — active agents present, others observe — reflecting real-world interdisciplinary oversight.

A "Pending Your Action" section surfaces items awaiting Dean approval (e.g., proposal approvals, module spec edits, MQA submissions) with inline document preview and editable drafts.

Curriculum-Market Alignment

Detailed curriculum-versus-market intelligence. A per-skill table shows Market Demand level, Programme Coverage percentage, and Gap indicator. AI-generated Key Insights panel highlights strengths and urgent gaps. Tabbed across Skills Alignment, Emerging Gaps, and Recommended Actions. Users can add a gap to the accreditation evidence pack directly from this page, which updates the Accreditation Hub's completeness state. The Roadmap section presents a structured timeline for curriculum reform including the TalentBank co-design partnership (RM 180K, CS5001 GenAI Systems module).

Student Readiness

Cohort-level skill progression monitoring. A colour-coded Skill Readiness Heatmap cross-tabulates Skills versus Year cohorts (High / Medium / Low / Not Started). An Intervention Queue lists at-risk students with assigned owners, deadlines, and status (Not Started / In Progress / Scheduled / Completed). Phase 2 adds three new urgent cases (Chris Lee — GPA drop; Priya Nair — attendance + mental health; Ahmed Al-Rashid — financial stress) to reflect narrative coherence with the AI Office Decision Room outcomes.

Accreditation Hub

Structured preparation for MQA, SETARA, QS, and internal reviews. Maps requirements to sources, owners, reporting periods, completeness status, and missing-item requests. Evidence packs from Curriculum-Market Alignment flow directly into the Hub, changing requirement completeness in real time via shared frontend state. Draft summaries are generated per requirement with clear ownership assignment.

Alumni Signal Intelligence

Graduate outcome tracking. KPI bar covers Employment Rate, Average Starting Salary, Average Time to First Job. Shows Graduate Destinations by role, Top Hiring Companies, Alumni Feedback, and an AI alumni-to-market insight analysis. Phase 2 adds GenAI/LLMs as the #1 skill gap (74% of Q2 2026 alumni survey respondents), directly feeding the Curriculum-Market Alignment evidence chain. Salary trend annotations show correlation between past curriculum changes and graduate salary improvements.

Collaboration Marketplace

Industry partnership discovery and management. Active partnerships now total 13, including TalentBank (new, RM 180K, GenAI co-design) and Grab (10 GenAI internship slots confirmed for Semester 6). Partnership health scores, event ROI, hiring conversion rates, and timeline histories are visible per partner. Recommended partner suggestions include Axiata (94% fit), Petronas Digital (88%), and CIMB (81%). Event creation and post-event impact reports complete the marketplace.

6.3 Employer Workspace

Persona: Edwin Khoo — HR Manager, Google Malaysia. Accessible at /employer.

Overview — Employer Intelligence Hub

Glassmorphism-styled command centre with a slim AI robot briefing row at the top. The left 25% is an NLP Chat panel — the employer types commands in natural language ("find me data science candidates with Python and cloud skills") and the system executes. The right 75% shows four intelligence cards: Pipeline Health, Pending Actions, Engagement Performance, and Market Talent Supply. The AI action queue flags urgent pipeline stalls and reactivation opportunities.

Talent Discovery

Primary candidate search and evaluation interface. Candidate Cards show name, skill tags, match percentage, and quick Save/Shortlist actions. AI-generated "Why It's a Strong Match" bullets reference verified evidence. Hiring Confidence Indicator, Recommended Action, and Key Risk to Verify help employers make fast, informed decisions. Evidence chips and Career Memory narrative are visible per candidate.

Campus Pipeline

Previous-candidate tracking and warm-talent reactivation. Surfaces candidates who participated in past challenges and events with current matching signals and context about when/where the employer met them. Urgent vacancy detection triggers pipeline alerts, allowing recruiters to reactivate prior strong candidates rather than restarting cold from a job board.

Create Engagement

Multi-step wizard for creating structured student engagement programmes. Supports Micro-Project, Workshop, Hackathon, and Mentorship engagement types. Live preview panel updates in real time. AI Engagement Strength Score explains talent attraction likelihood. Connects participation and performance to the Campus Pipeline for future candidate discovery.

Job and Internship Marketplace

Vacancy management and applicant review hub. KPI bar shows Active Positions, Total Applicants, Interviews Running, Positions Filled, and Average Match Rate. Applicant pipeline allows moving candidates across hiring stages with stage-specific AI briefing and suggested next actions per candidate.

7. Key Features

7.1 Student Features

- Career Readiness Score — live gauge of readiness based on skills and experience evidence.
- Career Memory — evidence-backed profile built from real experiences with conversational AI entry flow powered by Groq LLM.
- AI Skill Extraction — automatic extraction of skills and confidence scores from experience entries via real Groq inference.
- NLP Navigation — natural language intent classification routes users to the correct page and pre-fills actions.
- Career Intelligence — tabbed view of skill clusters, career path matches (Career Graph), and market insights.
- Opportunity Matching — events and jobs ranked by AI match percentage against student profile.
- Application Pipeline Tracker — Kanban board across five hiring stages.
- Interview Practice — repeatable mock sessions with application-aware prompts and feedback.
- AI Career Companion — primary interface providing proactive briefings, quick reply chips, and conversational guidance.
- Communities & Mentorship — peer support, portfolio feedback, career-switch stories, and mentor discovery.

7.2 University Features

- University AI Office — five specialised AI agents (Dr. Atlas, Prof. Nova, Dean Echo, Dr. Iris, Aria) with domain briefings and Decision Room.
- Decision Room — structured modal where agents present recommendations and the Dean provides human approval.
- Pending Your Action — inline approval queue for proposals, module specs, intervention plans, and compliance filings.
- Programme-Market Alignment — per-skill comparison of curriculum coverage versus employer demand with evidence pack integration.
- Curriculum Roadmap — structured reform timeline with partner co-design milestones (TalentBank, Grab).
- Student Readiness Intervention Queue — at-risk detection with owner, deadline, and status tracking.
- Accreditation Hub — MQA/SETARA/QS requirement mapping with evidence reuse across frameworks.
- Alumni Signal Intelligence — graduate outcomes, salary trends, role pathways, and feedback routing.
- Collaboration Marketplace — partnership portfolio (13 active), event creation, and post-event impact reports.
- Cross-Page State — interventions, partnerships, and evidence packs update KPIs across all pages via Zustand.

7.3 Employer Features

- Evidence-Based Talent Discovery — candidate search filtered by verified skills and match percentage.
- AI Match Explainability — bullet-point reasons why a candidate fits, with evidence references.
- Hiring Confidence Indicator — AI-generated confidence score and key risk flags per candidate.
- Campus Pipeline — warm-talent history and urgency-driven reactivation from past events.
- NLP Employer Chat — natural language command panel driving talent search and pipeline actions.
- Create Engagement Wizard — multi-step builder for Micro-Projects, Workshops, Hackathons, and Mentorship pods.
- Job and Internship Marketplace — vacancy posting, applicant tracking, and suggested candidate surfacing.

8. Technical Architecture

CareerOS Phase 2 is built on a modern frontend-first stack with real LLM integration via the Groq API. All state management is handled client-side using Zustand. The architecture is designed for prototype speed while maintaining a clear path to production scalability.

8.1 Tech Stack Overview

Layer	Technology	Role
Frontend	React + Vite	Three workspace SPAs with shared component library
Styling	Tailwind CSS	Utility-first responsive design system
Routing	React Router v6	Workspace routing and page navigation
State	Zustand	Cross-page mock state — interventions, partnerships, evidence packs
Icons	lucide-react	Consistent icon system across all workspaces
LLM	Groq API (Llama 3)	NLP navigation, skill extraction, AI companion responses
Backend	FastAPI (partial)	Candidate chat and add-experience routes (prototype)

8.2 AI Architecture — Groq + NLP Layer

Phase 2 introduces real LLM inference via the Groq API (model: llama3-8b-8192). Groq's free tier provides fast, low-latency responses suitable for hackathon demo conditions. The NLP layer handles two primary functions:

- **Intent Classification (NLP Navigation):** user types a natural language command → sent to Groq with an intent classification system prompt → returns JSON: { intent, destination, action, prefill } → frontend navigates and pre-fills.
- **Skill Extraction (Career Memory):** user describes an experience in free text → sent to Groq with a skill extraction prompt → returns detected skills with evidence context → displayed as confirmation card with "Skills detected by AI" label.

The Groq API key is stored in a .env file as VITE_GROQ_API_KEY. All other AI capabilities in the prototype use scripted fallbacks with pre-defined outputs; only Career Memory skill extraction and NLP navigation use live Groq inference.

8.3 Infrastructure

Component	Detail
Frontend Hosting	Vercel — continuous deployment from GitHub, instant CDN distribution
Development Server	Vite dev server at localhost:5173

State Management	Zustand — in-memory, no backend required for demo
API Key Management	.env file (VITE_GROQ_API_KEY) — not committed to repository
Backend (partial)	FastAPI (Python) — candidate and employer route prototypes

8.4 Data Sources

- Groq API (Llama 3 8B) — live LLM inference for NLP navigation and skill extraction.
- Mock data — all prototype demo data is simulated; no real institutional or employer data is connected in the current prototype.
- Zustand stores — cross-page state for interventions, partnerships, evidence packs, and accreditation overrides.

9. AI and Intelligence Layer

Phase 2 introduces real LLM inference at two critical points (NLP Navigation and Skill Extraction). All other AI capabilities use scripted simulation with pre-defined outputs. The architecture is designed for real AI backend integration in the next development phase.

Capability	Input	Output	Status
NLP Navigation	Natural language user intent	JSON: intent, destination, action, prefill	Live — Groq API
Career Memory Skill Extraction	Student experience description (free text)	Extracted skills, confidence scores, evidence traces	Live — Groq API
AI Companion Briefing	Student profile, recent activity, deadlines	Proactive briefing, signal cards, quick reply chips	Simulated
Career Intelligence	Student skills, target roles, market data	Path recommendations, skill gaps, salary benchmarks	Simulated
Opportunity Matching	Student profile, event and job metadata	Ranked opportunities with match badges	Simulated
Programme Alignment Engine	Programme skills, market demand values, gap data	Alignment tables, gap spotlight, recommended actions	Simulated
University AI Office Agents	Department-specific data + user topic keyword	Agent briefings, Decision Room recommendations	Simulated + optional Groq
Candidate Match Advisor	Candidate profile, evidence, match score	Fit reasons, risk flags, recommended next step	Simulated
AI Engagement Strategist	Engagement type, goals, target audience	Programme recommendations, projected outcomes	Simulated

10. Market Opportunity

CareerOS targets the intersection of three large and growing markets: HR technology, EdTech, and talent intelligence platforms.

- 85% of graduates are reported to be underqualified for available roles at the point of hiring.
- The global skills mismatch is estimated to cost the economy USD 2.4 trillion annually in lost productivity.
- The average job search for a fresh graduate takes over six months.
- 72% of employers report difficulty evaluating early talent from traditional resumes and interviews alone.

- Global HR technology market: projected to exceed USD 40 billion by 2027.
- Southeast Asia EdTech market: one of the fastest-growing in the world.
- University career services: underdigitised and underserved across Southeast Asia.

Generative AI has made it feasible to extract skills from unstructured experience data, explain candidate fit in natural language, and surface programme gaps from labour market signals — all at a cost structure that was not viable two years ago. CareerOS is positioned to bring this capability to higher education institutions in Southeast Asia at the moment of peak readiness.

11. Competitive Landscape

CareerOS occupies a unique position in the talent and career technology market. No existing platform simultaneously serves students, universities, and employers within a shared evidence and intelligence layer.

Platform	Primary Focus	Key Limitation	CareerOS Advantage
LinkedIn	Professional networking	Unverified, keyword-based profiles	Evidence-first, readiness-focused, institutional layer
Handshake	Campus recruiting	No university intelligence layer	Programme alignment, intervention, event impact
Indeed	High-volume job search	Listing-first, no readiness system	Career preparation and evidence building
Glassdoor	Company reviews and salary	Retrospective; no preparation tools	Forward-looking readiness intelligence
JobStreet	SEA job listings	No institutional or evidence layer	Unified three-side ecosystem
Hiredly	Malaysian Gen Y/Z job board	Personality matching, no university workspace	Verified evidence; institutional intelligence
LMS Platforms	Course content delivery	No employer connection or market signal	Employability outcomes, market alignment

CareerOS's strategic whitespace: the only platform that is simultaneously evidence-first and multi-stakeholder, serving all three sides of the career readiness ecosystem within a connected intelligence layer. Phase 2's AI-first OS positioning further extends this gap — no competitor offers an NLP-navigated, action-first interface for both students and institutional leaders.

12. Business Model

CareerOS operates a multi-sided SaaS model with differentiated revenue streams per stakeholder group.

12.1 Revenue Streams

- **University SaaS subscription** — annual or semester-based licence per institution, including AI Office, Student Readiness, Curriculum Alignment, and Accreditation Hub.
- **Employer subscription** — monthly or annual access to Talent Discovery, Campus Pipeline, job marketplace, and engagement wizard.
- **Premium analytics** — advanced cohort benchmarking, cross-programme comparison, and predictive readiness modelling.
- **Event impact reporting** — packaged post-event reports sold following industry-campus collaboration events.
- **Managed engagement programmes** — employer-funded challenges, workshops, and hackathons delivered through the CareerOS platform with student evidence capture.

12.2 Go-to-Market Approach

CareerOS enters through university partnerships, leveraging institutional relationships to onboard both students and employers as secondary stakeholders. This mirrors the Handshake playbook adapted for the Southeast Asian higher education context, with the university as the primary anchor customer.

Recommended first wedge: a faculty-level graduate-readiness intervention pilot — Career Memory for one cohort, readiness-risk detection, one employer-backed challenge or workshop, and evidence of actions/outcomes. This wedge has an accountable buyer, bounded data, visible student benefit, and measurable operational outcomes.

13. Product Roadmap

Phase 1 — Complete (Hackathon Round 1)

Dashboard-first prototype with three workspaces, mock AI, and no real LLM integration.

- Student Workspace: Overview, Career Memory (form), Career Intelligence, Opportunities, Applications.
- University Workspace: Overview, Programme-Market Alignment, Alumni Signals, Student Readiness, Collaboration Marketplace.
- Employer Workspace: Talent Discovery, Create Engagement Wizard, Job and Internship Marketplace.
- All AI capabilities simulated with mock data and pre-defined outputs.

Phase 2 — Complete (Hackathon Round 2 · July 2026)

AI-first OS evolution with real Groq LLM integration, NLP navigation, and enhanced University workspace.

- NLP Navigation Layer — natural language intent routing to any workspace/module via Groq API.
- Conversational Career Memory — step-by-step experience entry with real Groq skill extraction.
- AI Companion as primary interface — proactive briefings, quick reply chips, robot mascot UX.
- University AI Office — five specialised AI agents, Decision Room with all-agent participation, Pending Your Action queue.
- Accreditation Hub — MQA/SETARA/QS requirement mapping with cross-page evidence reuse.
- Narratively coherent mock data — TalentBank partnership, Grab GenAI slots, Chris/Priya/Ahmed interventions, July 2026 timeline.
- Employer glassmorphism redesign — NLP chat panel, slim AI briefing row, four intelligence cards.

Phase 3 — Next 60 Days

- Connect real AI backend using Google ADK — activate a four-agent orchestration system.
- Integrate O*NET Web Services and Lightcast Open Skills Taxonomy for live skill and market demand data.
- Deploy production backend on Google Cloud Run with Cloud SQL (PostgreSQL) and Secret Manager.
- Live programme-market alignment using real employer skill demand signals.
- Google and LinkedIn OAuth for role-based user sign-on.

Phase 4 — 6 to 12 Month Vision

- Institutional benchmarking — compare readiness and market alignment across faculties and programmes.
- Real university SIS and LMS integration — ingest syllabus and course outlines to map skill coverage automatically.
- Mentor and network features — mentor matching, network graph, and advisory session scheduling.

- Verified challenge-to-Memory-to-hire propagation — employer challenge performance flows into student evidence and employer candidate profiles.
- Continuous accreditation assembly across MQA, SETARA, QS, and internal reviews.
- CareerOS becomes the career readiness operating system for universities, students, and early talent employers in Southeast Asia.

14. Team

Team GongBaiWan — TalentBank Hackathon 2026

A multidisciplinary team combining product design, frontend engineering, AI integration, and business strategy — united by a shared conviction that career outcomes deteriorate when evidence, confidence, relationships, and institutional action decay between isolated moments. CareerOS is our response to that decay.

15. Setup and Demo Instructions

15.1 Running the Frontend Prototype Locally

- Clone the repository: git clone https://github.com/Kinston-Chee/Career_OS.git
- Navigate to the project directory: cd Career_OS
- Install dependencies: npm install
- Create a .env file in the project root and add your Groq API key: VITE_GROQ_API_KEY=your_key_here
- Start the development server: npm run dev
- Open your browser at the local development URL (typically http://localhost:5173)
- Navigate to /student, /university, or /employer to access each workspace.

15.2 Groq API Key

The Groq API (free tier) is required for NLP Navigation and Career Memory skill extraction. Obtain a free key at console.groq.com. Without a key, these features will fall back to scripted demo responses. All other features work fully offline with mock data.

15.3 Current Frontend Deployment

Visit the live deployment to view the Phase 2 frontend prototype. Complete backend functionality will be added in Phase 3.

15.4 Recommended Demo Flow

- **Student:** Overview briefing → NLP: "I want to add my internship experience" → Career Memory conversational flow with live skill extraction → Career Intelligence graph → Opportunity matching.
- **University:** AI Office → Enter Decision Room (TalentBank partnership) → Curriculum Alignment (add GenAI to evidence pack) → Accreditation Hub (see completeness update) → Student Readiness (view intervention queue).
- **Employer:** Overview NLP chat ("find GenAI candidates") → Talent Discovery (evidence-rich candidate profiles) → Campus Pipeline (warm reactivation) → Create Engagement Wizard.

16. Appendix

16.1 Terminology Glossary

Term	Definition
CareerOS	The product name — the AI-powered career operating system.
CareerGraph	The intelligence layer connecting skills, evidence, opportunities, employers, and university outcomes across all three workspaces.
Career Memory	The student evidence profile built from experiences, skills, and verification signals — now populated via conversational AI flow.
NLP Navigation	The Phase 2 natural language routing layer — user intent is classified by Groq and translated into workspace navigation and pre-filled actions.
University AI Office	The institutional hub of five specialised AI agents (Dr. Atlas, Prof. Nova, Dean Echo, Dr. Iris, Aria) that monitor domains and surface Decision Room briefings.
Decision Room	A structured modal within the AI Office where an agent presents a prepared recommendation and the Dean provides human approval.
Programme-Market Alignment	University-facing analysis comparing programme skill coverage with employer market demand.
Accreditation Hub	Structured preparation for MQA, SETARA, QS, and internal reviews with evidence reuse and ownership tracking.
Groq API	The LLM provider used in CareerOS (model: llama3-8b-8192). Free tier, fast inference, no local installation needed.
Zustand	The client-side state management library used for cross-page state (interventions, partnerships, evidence packs) without a backend.
CareerGraph	The product's core intelligence layer — also the name of the interactive career path exploration UI in the Student workspace.
TalentBank	The hackathon organiser and strategic partner — also a new curriculum co-design partner in the mock data (RM 180K, CS5001 GenAI Systems module).

16.2 Phase 2 Evidence Status

Claim	Status
Three working workspaces (Student, University, Employer)	Repository-demonstrated
NLP Navigation via Groq API	Live — Groq API (llama3-8b-8192)
Career Memory skill extraction via Groq API	Live — Groq API (llama3-8b-8192)

University AI Office with Decision Room	Repository-demonstrated interaction model; scripted fallback + optional Groq
Cross-page Zustand state (interventions, partnerships, evidence packs)	Repository-demonstrated
Accreditation Hub evidence reuse (curriculum pack → requirement completeness)	Repository-demonstrated frontend state handoff
Verified candidate evidence	Interface models verification; production provenance not complete
Autonomous AI agents	Specialised assistant interaction model with human approval; autonomy is future capability
Live institutional/employer data	All demo data is simulated mock data
Measured outcomes or impact	Not yet measured; pilot metrics defined for future phase